PHYC10009 Physics Foundations

Nuclear Fusion, Power for civilisation
Week 12 Lecture 2

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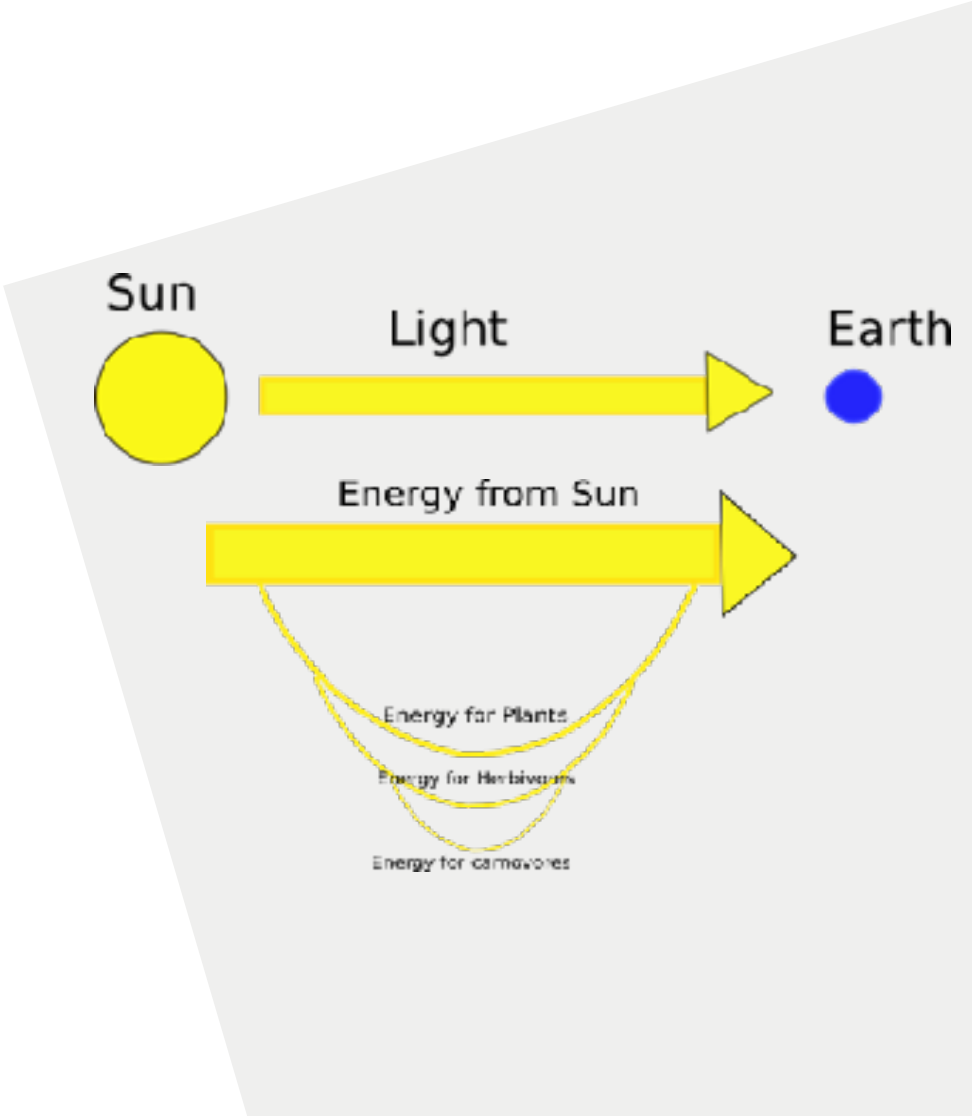
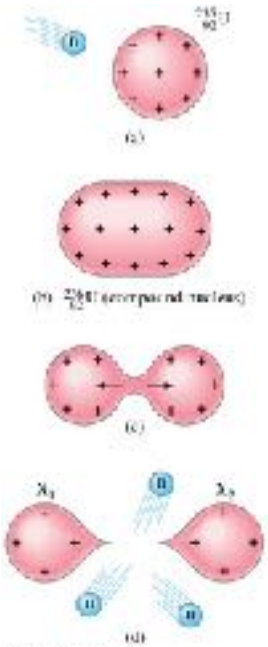
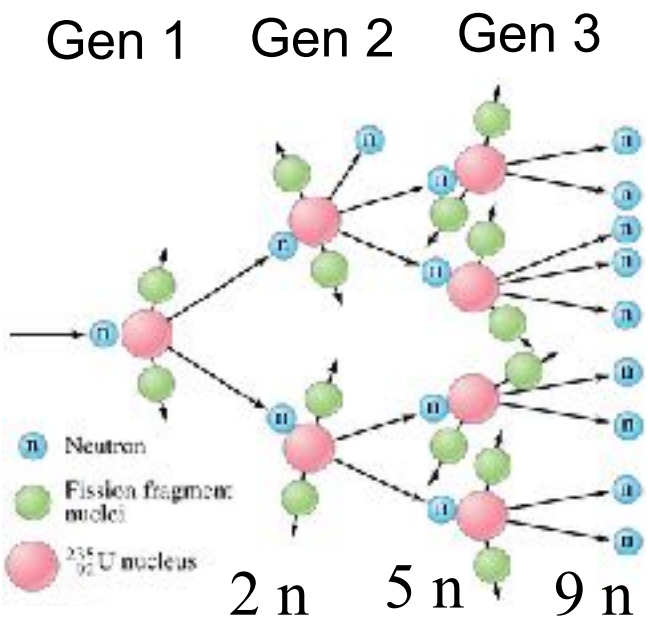


Reminder

Nuclear Reactions, Nuclear Fission



Applications of Nuclear fission:
Nuclear chain-reactions



$N = N_0 k^n$, Time = $n \times R_t$

Exponential Growth
(Covid-19)

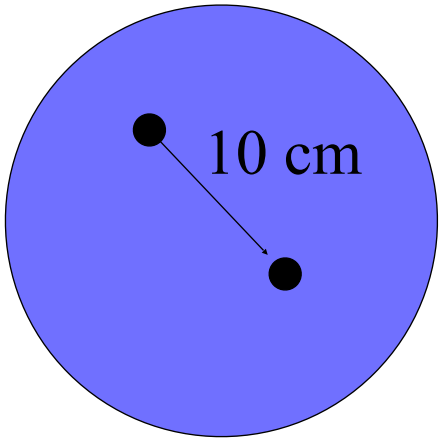
Examples - 1

Calculate the speed of a 1 MeV neutron typical of a fission reaction.

$$\text{K.E.} = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{2 \times 1 \times 10^6 \times 1.6 \times 10^{-19}}{1.67 \times 10^{-27}}} = 1.4 \times 10^7 ms^{-1}$$

A neutron travels typically 10 cm in ^{235}U before initiating a fission reaction. Calculate time between successive fission reactions.

$$R_t = \frac{x}{v} = \frac{0.1}{1.4 \times 10^7} = 7.1 \times 10^{-9}\ s = 7\ ns$$



Examples - 2

Assume the multiplication factor, k, in a sphere of ^{235}U of radius 10 cm is k=1.5. The number of nuclei in the sphere is 5×10^{27}, estimate the number of generations, and hence the time required to consume half of them from a single free neutron.

Step one. Calculate the number of generations required to consume:

$$\frac{5 \times 10^{27}}{2} = 2.5 \times 10^{27} \text{ nuclei}$$

Use: $N(n) = 1 \times (k)^n = 2.5 \times 10^{27}$

So: $\log_e(k^n) = \log_e(2.5 \times 10^{27}) \Rightarrow n\log_e\left(1.5\right) = 63.1$

$$\Rightarrow n = \frac{63.1}{0.41} = 154$$

Time taken for this: $t = n \times R_t = n \times 7 \times 10^{-9} = 1.1 \times 10^{-6}\ s = 1.1\mu s$

So **1 microsecond to consume half the nuclei**

Examples - 3

Starting a light water nuclear reactor, the operators adjust the multiplication factor, k, to be k = 1.001. The time between each generation is set by neutrons released after beta-decays of fission products. On average, this is $R_t = 8$ seconds.

Calculate the time taken to double the power output of the reactor.

$$P(n) = P_0\,(k)^n$$

Want $k^n = 2 \Rightarrow \log_e\left(k^n\right) = \log_e 2 \Rightarrow n\log_e(k) = \log_e 2$

$$\Rightarrow n = \frac{\log_e 2}{\log_e 1.001} = \frac{0.693}{0.00099} = 693 \text{ generations}$$

Since each generation takes 8 seconds, that is $693 \times 8 = 5500$ seconds or **1.5 hours**

Covid-19

The generation time for Covid-19 appears to about 7 days, so $R_t = 7$ days. In New York City, before social distancing, the multiplication factor k was about 3.

Starting from a single infection, estimate the time taken to infect 2 million people.

Covid-19

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Starting from a single infection, estimate the time taken to infect 2 million people.

Let N= number of infections. $n =$ number of generations.

$$N(n) = N_0 k^n = 2 \times 10^6 \Rightarrow \log_e k^n = \log_e(2 \times 10^6)$$

$$\Rightarrow n\log_e(3) = \log_e(2 \times 10^6) \Rightarrow \mathrm{n} = \frac{\log_e 2 \times 10^6}{\log_e 3} = \frac{14.5}{1.1} = 13 \text{ generations}$$

The time taken for this is: $13 \times R_t = 13 \times 7 = 91$ days.

ie **about 3 months**. (January, 2020 to April 2020)

The extreme social distancing in place in NYC (and Australia) put $k \sim 0.75$

Covid-19

After implementing social distancing, NYC, reduced k from 3 to 0.75, calculate the time taken to reduce the average daily rate of infections from 20,000 per day to 1,000 per day.

$$R(n) = R_0(k)^n \Rightarrow \log_e R(n) = \log_e(R_0(k)^n) = \log_e(R_0) + nlog_e(k)$$

$$\Rightarrow \log_e 1000 = \log_e 20000 + nlog_e(k)$$

$$\Rightarrow n = \frac{\log_e(1000) - \log_e(20000)}{\log_e(0.75)} = \frac{6.907 - 9.903}{-0.2877} = 10.41$$

Time = $R_t \times n = 7 \times 10.41 = 73\ days$

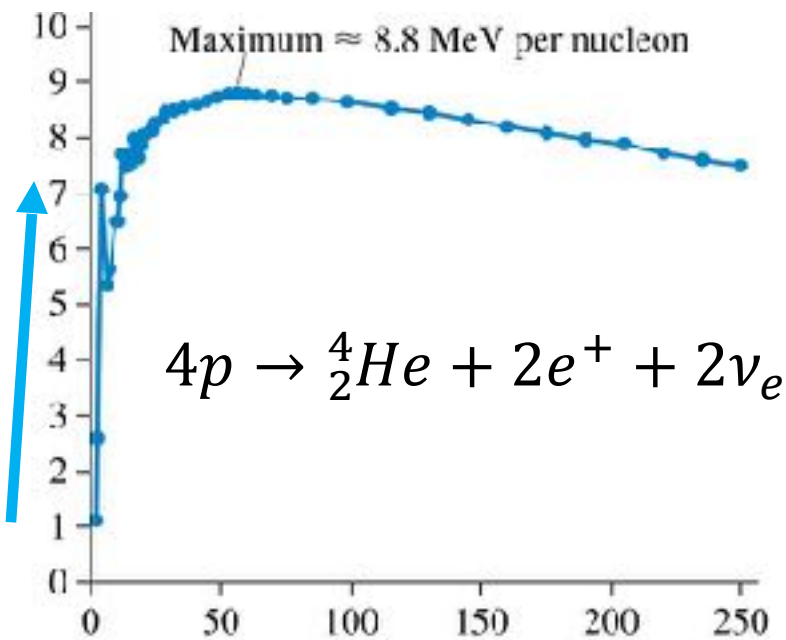
Nuclear Fusion G31.3

The lightest nuclei can fuse to form heavier nuclei, releasing energy in the process. An example is the sequence of fusion processes that change hydrogen into helium in the Sun. They are listed here with the energy released in each:

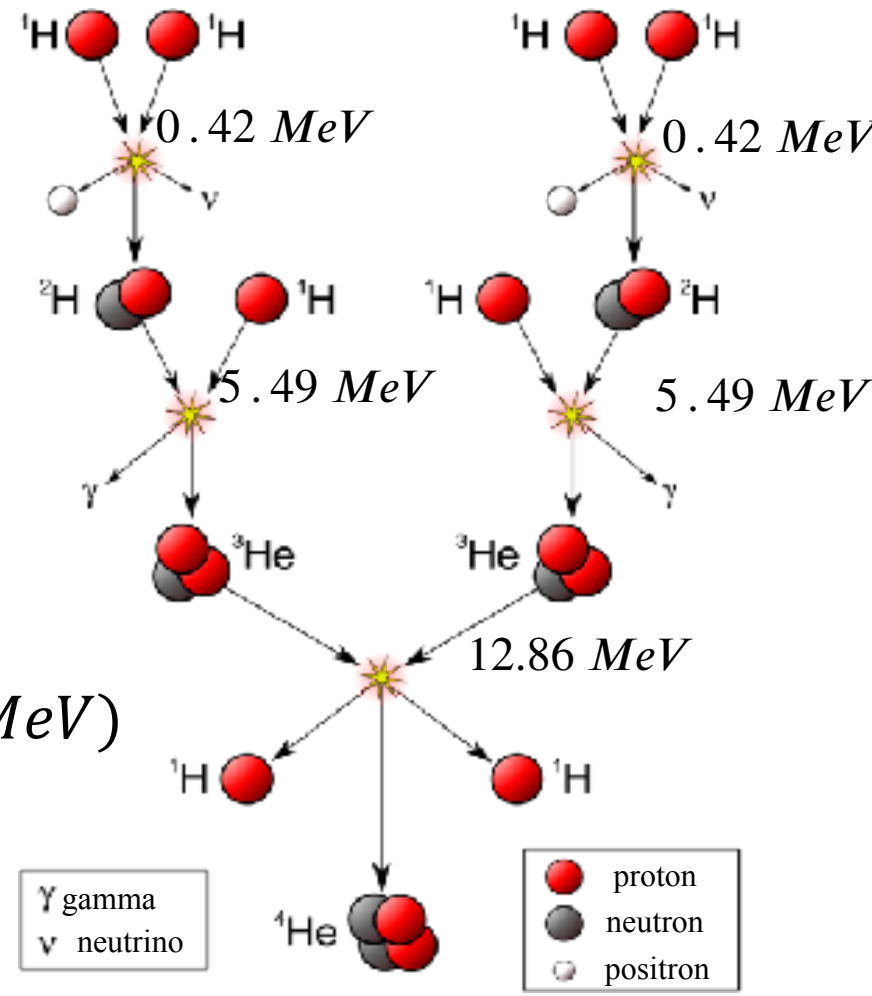
$$^{1}_{1}H + ^{1}_{1}H \rightarrow ^{2}_{1}H + e^{+} + \nu_e \quad (0.42\ MeV)$$

$$^{1}_{1}H + ^{2}_{1}H \rightarrow ^{3}_{2}He + \gamma\ (5.49\ MeV)$$

$$^{3}_{2}He + ^{3}_{2}He \rightarrow ^{4}_{2}He + ^{1}_{1}H + ^{1}_{1}H \quad (12.86\ MeV)$$



$$4p \rightarrow ^{4}_{2}He + 2e^{+} + 2\nu_e\ (24.7\ MeV)$$



Human use of Nuclear fusion

The reactions listed below are being investigated as a source of power for humans

$$ {}^{2}_{1}H + {}^{2}_{1}H \rightarrow {}^{3}_{1}H + p \quad (4.03 MeV) $$

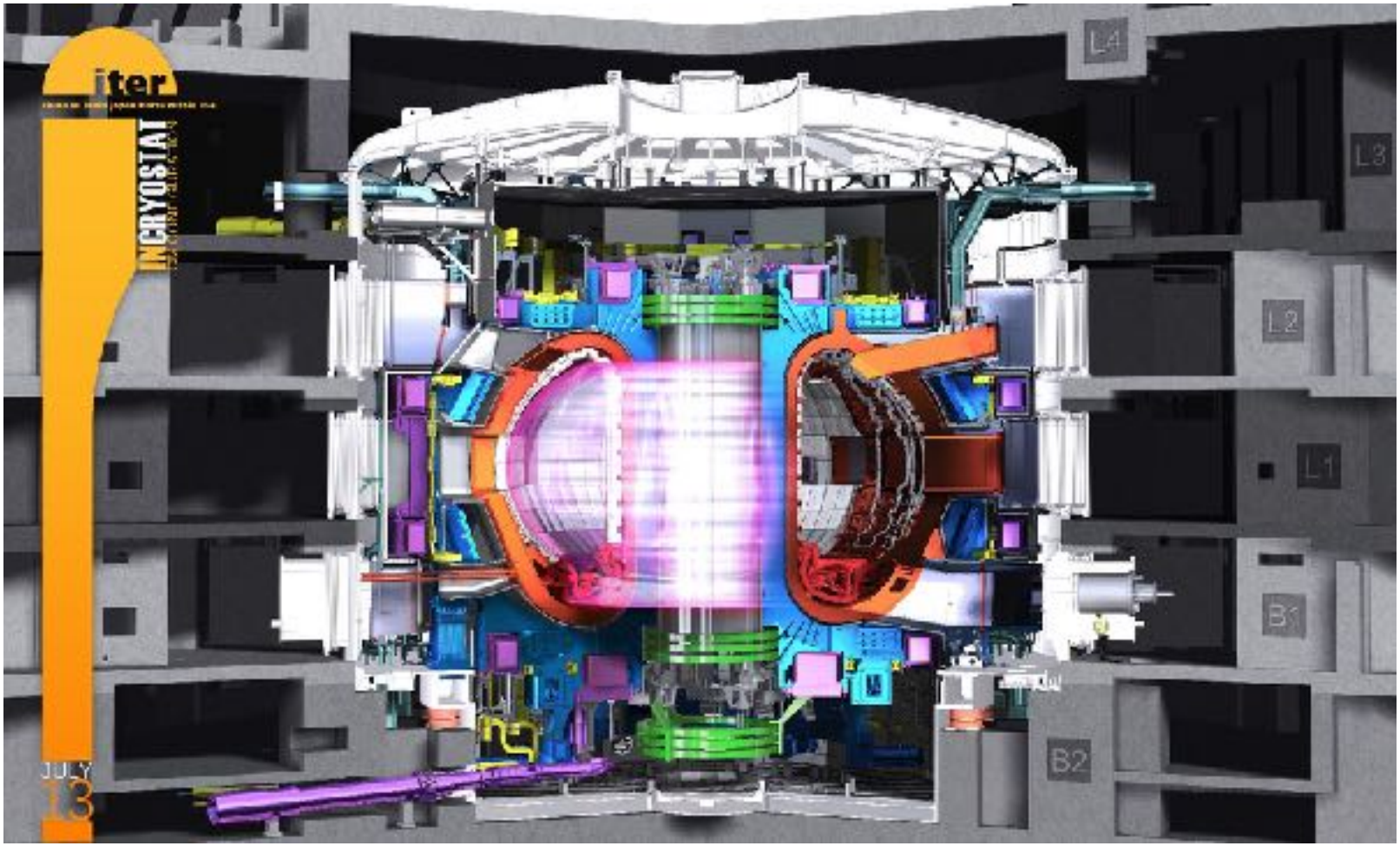
$$ {}^{2}_{1}H + {}^{2}_{1}H \rightarrow {}^{3}_{2}He + n \quad (3.27\ MeV) $$

$$ {}^{2}_{1}H + {}^{3}_{1}H \rightarrow {}^{4}_{2}He + n \quad (17.59\ MeV) $$

- These processes have significant safety advantages over Nuclear Fission but building a power reactor that employs these has not yet been achieved.
- The Coulomb repulsion between the nuclei make it very hard to make macroscopically large numbers of reactions.
- Need temperatures above 100 M degrees for a sustained period of time.

ITER

ITER is an international collaboration to build a Nuclear Fusion device close to what is needed for a commercial Power Plant. The current cost of the project is $20 Billion. It will heat a plasma to 150 million degrees.



http://iter.org

Energy and Power

This rest of this lecture is almost all qualitative and not expected to be memorised

- Energy appears consistently throughout this course
- Originally in Mechanical form
- Then as transmitted in Sound and waves
- Electrical form
- Photons
- Atomic structure
- Nuclear structure and transformations
- Ionising radiation

Energy is never created or destroyed
It can be transformed from one form to another

Power is the rate of transformation of energy

$$P = \frac{dE}{dt} \; J/sec \text{ or watts}$$

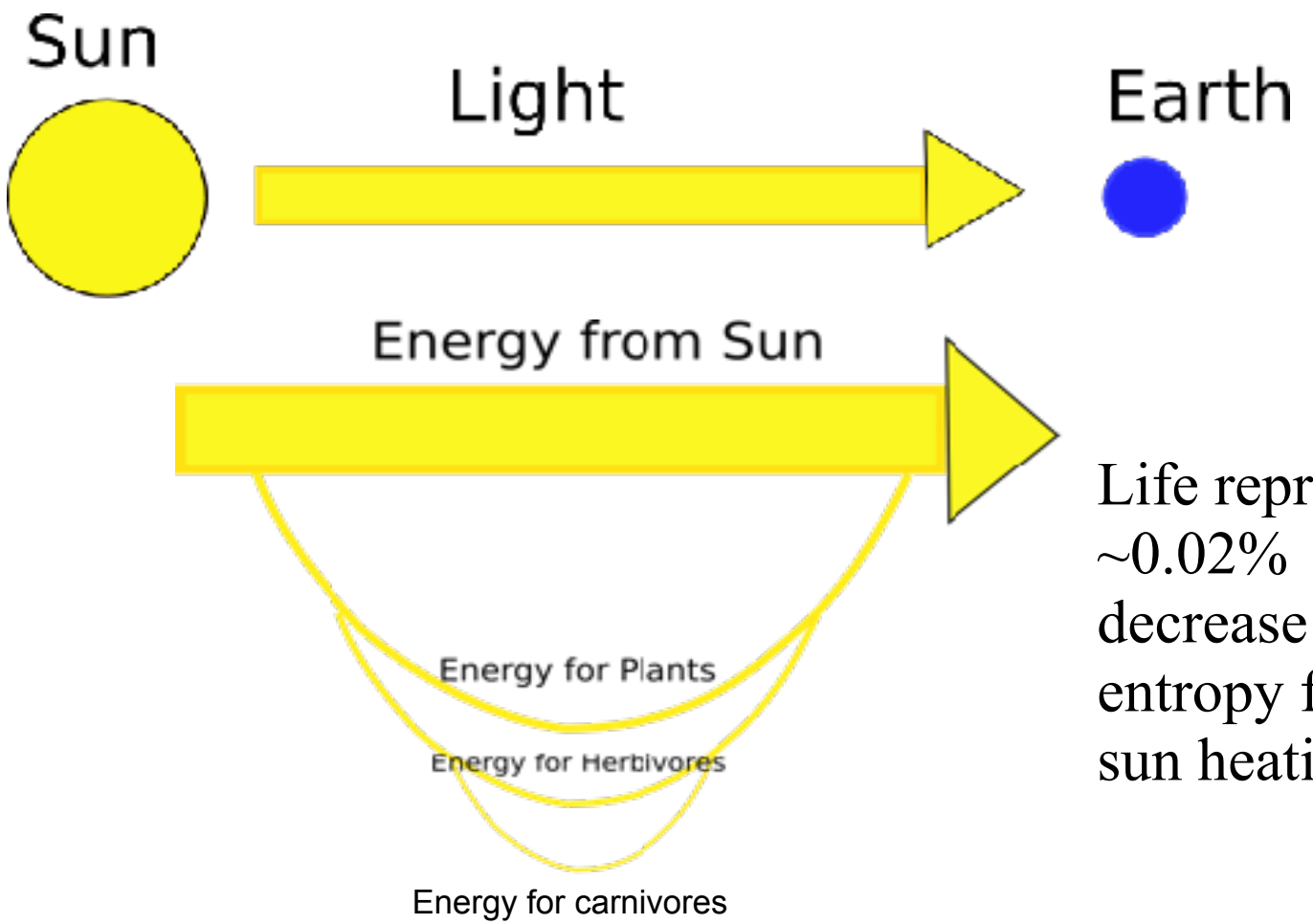
Energy and Entropy G15.7

- The entropy of a state is a measure of its likelihood. The larger the entropy the more likely it is.
- 2nd Law of Thermodynamics:

"All energy transformations increase Universal entropy."

- Life is in a very low state of entropy, "S"
- To exist it must create large amounts of entropy elsewhere.
- (S = Q/T)
- S = Energy Flow/temperature
- Life requires large amounts of Energy flow.
- Energy flow == Power

Life and Power



Life represents a ~0.02% decrease in entropy from the sun heating earth

Energy and Power for our Civilisation

A healthy human can sustain an average daily power output of ~30 watts.
One human energy over one year = $30 \times 3600 \times 24 \times 365 \sim 1 \times 10^9 J$
Australian Energy consumption = $6.2 \times 10^{18}\ J = 250\ GJ/person$

So on average every person consumes enough power equivalent to the efforts of 250 people.

This is the basis of our civilisation and our comfortable life.
We use Power for:

- Transportation: Cars, trains, trucks, aircraft, ships etc.
- Industry: Making all the stuff we use – particularly mining, steel and concrete
- Agriculture: we create food in vast abundance
- Electricity: Internet, computers, lighting,…….

Energy and civilisation

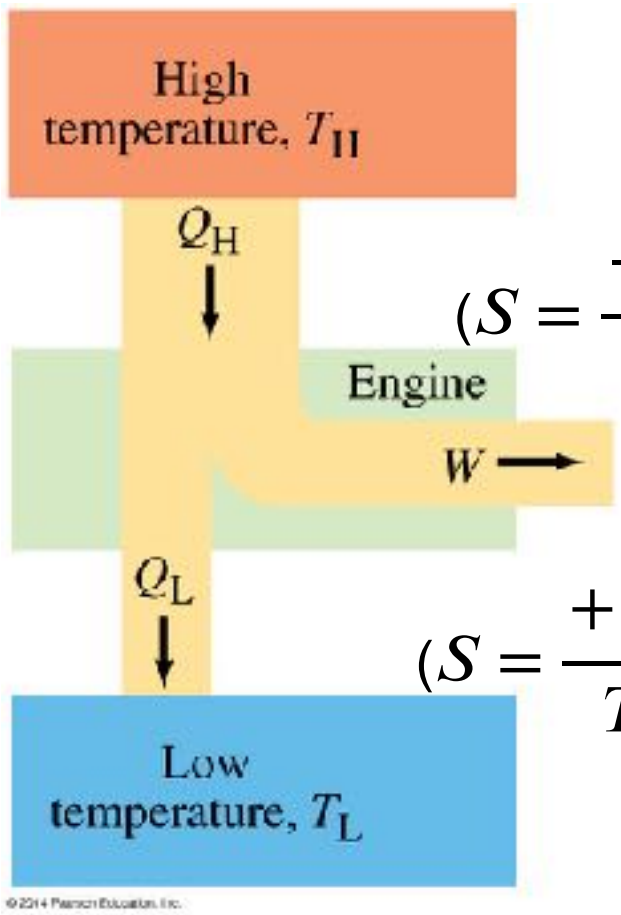
- Our Civilisation is based on cheap energy and machines
- Previous civilisations utilised humans and animals. (Still the case for large parts of the world.)
- Given sufficient quantities of energy our civilisation can generate all the products it needs. (Food, Health, Metals, Plastics, Water)
- The vast use of Energy by our Civilisation is a manifestation of creating regions of low entropy.

Energy vs Power

- Suppose you receive $7.6 Million in your Bank Account- Woo Hoo!
- Suppose you receive $7.6 Million but placed $1 in each square km of Australia
  - Not so great.
- Same amount of money but nowhere near as useful.
- What civilisation needs (and will spend money on) is Power
- How much energy you can transform per Second is Power

Heat Engines and mechanical power - G15.5

Heat engines turn heat flows into mechanical power



High Temperature source of heat

$$(S = \frac{-Q_H}{T_H}, \text{ Entropy decreases})$$

Mechanical Power that can do work

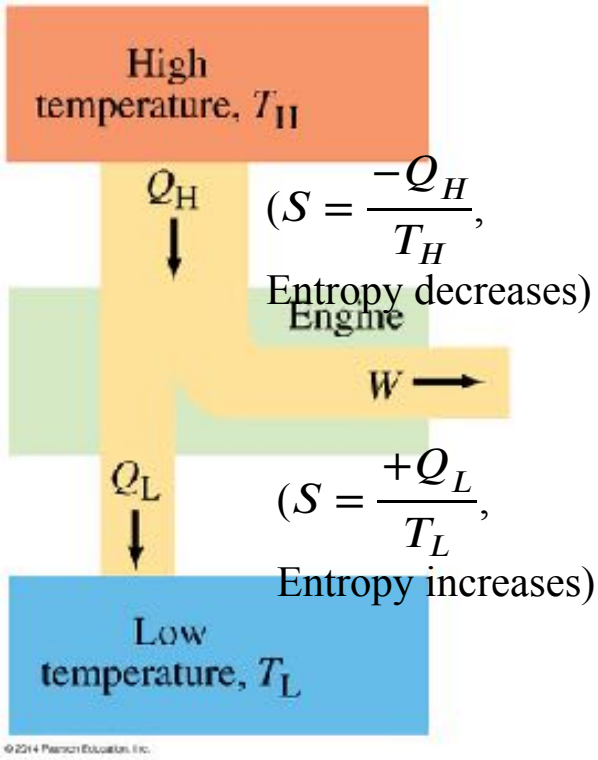
$$(S = \frac{+Q_L}{T_L}, \text{ Entropy increases})$$

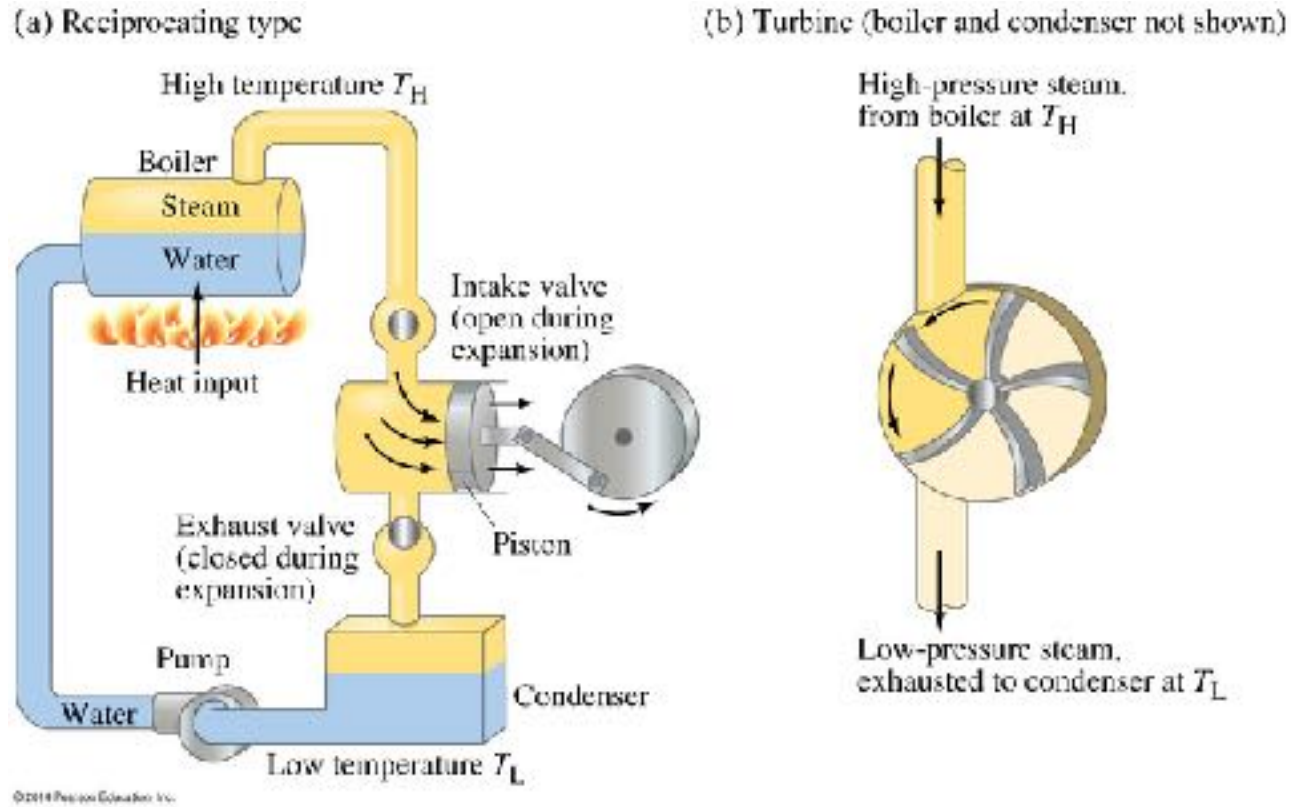
Low Temperature sink of heat

Require: $\frac{+Q_L}{T_L} + \frac{-Q_H}{T_H} > 0$

Heat Engines and mechanical power

Heat engines turn heat flows into mechanical power



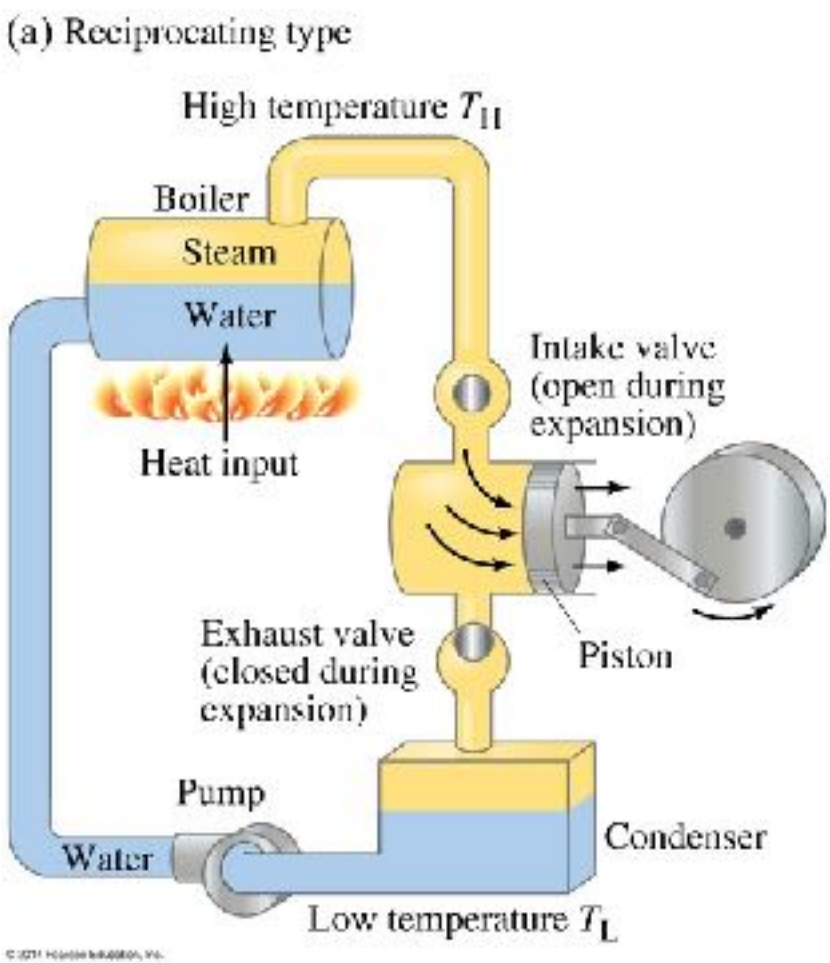


Not all the energy in the heat flow can be converted into mechanical power. 2nd law of thermodynamics limits the efficiency, $\mathcal{E}$, to:

$$\mathcal{E} = \frac{T_H - T_L}{T_H}$$

Temperature in units of Kelvin

Example:



A high pressure steam engine heats its boiler and injects steam at 200 degrees Celsius into a piston.

The piston does mechanical work and exhaust the steam into a condenser at 50 degrees Celsius.

What is the maximum efficiency of the engine?

$$\text{Max } \mathscr{E} = \frac{T_H - T_L}{T_H}$$

$$T_H = 200 + 273 = 473K$$

$$T_L = 50 + 273 = 323K$$

$$\Rightarrow \mathscr{E} = \frac{473 - 323}{473} = 0.32$$

The actual efficiency will be much smaller

Direct use of Mechanical Power

Mechanical power is used directly for transportation:
Aircraft, cars, trains, ships etc.
The source of energy required for their power are almost always liquid fossil fuels: Gasoline, Kerosene, Diesel

These are easily transported and have a high energy density.

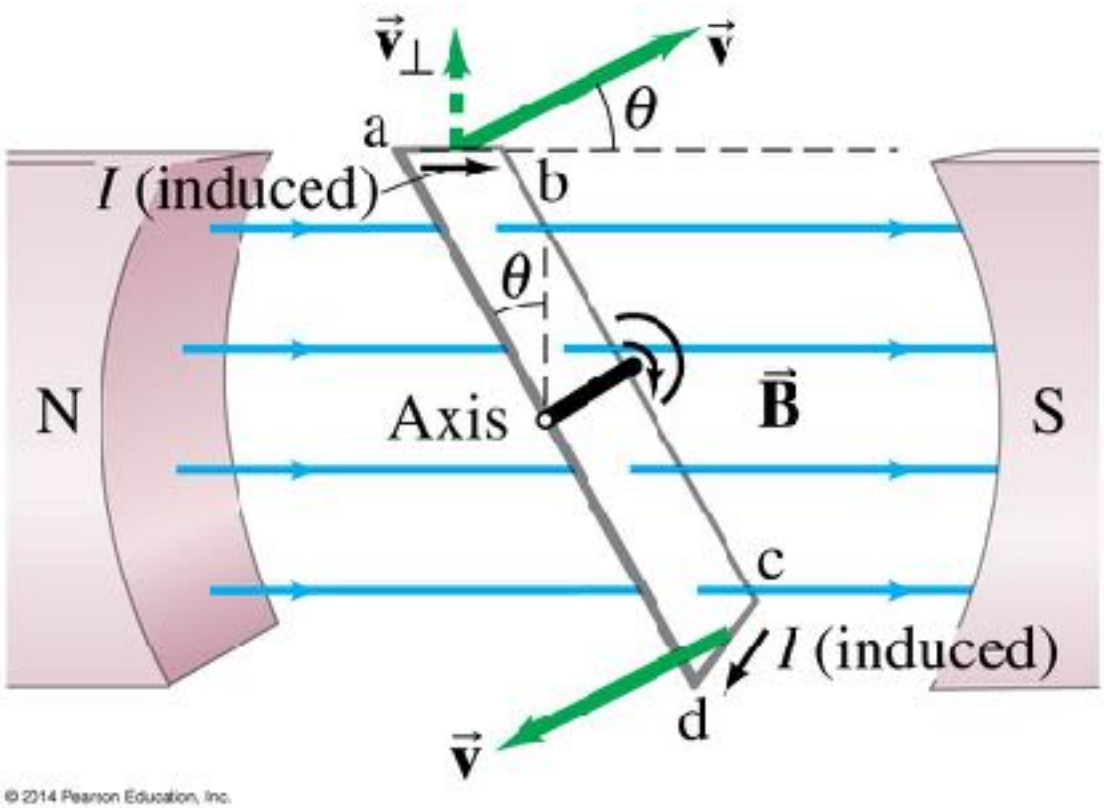
Gasoline: Energy/kg = 46.7 MJ/kg or 13 kWh/kg

The vast majority of these liquid fuels are sourced from crude oil.

The efficiency of these motors that directly convert chemical energy to mechanical power is in the range 20% - 50%.

Mechanical Power and electricity G21.5

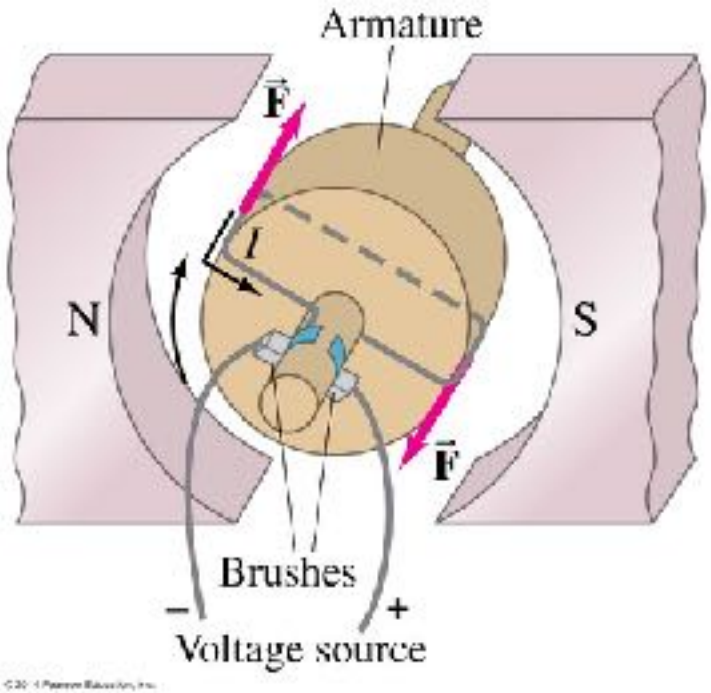
- Mechanical Power and Electrical Power can be interchanged with high efficiency.
- Mechanical Power is transformed into Electrical Power via generators.
- These rotate wire coils in a magnetic field to generate an electrical current.



Highly refined designs have efficiencies of around 98%

Mechanical Power and electricity G20.10

Electrical power can be converted to mechanical power via electric motors. These are artfully arranged wire coils that generate magnetic fields that are alternatively attracted and repelled by an external magnetic field to produce a mechanical rotation.



There are many different designs for different applications.

A refined design can reach 98% efficiency.

Power for our civilisation

The vast majority of power employed by our civilization comes from burning Hydrocarbons in Oxygen: Oil, Coal and Natural Gas.

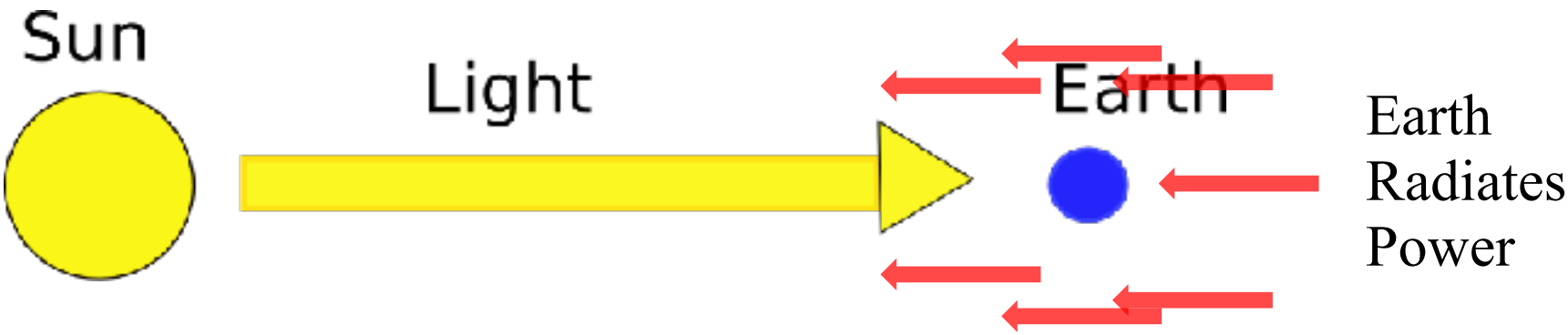
The Chemical Reactions are something like:

$$C_nH_m + O_2 \rightarrow n\ CO_2 + \frac{m}{2} H_2O$$

Since the Oil, Coal and Natural Gas are sourced from biological activity hundreds of million years in the past, the effect is to release CO_2 into the atmosphere.

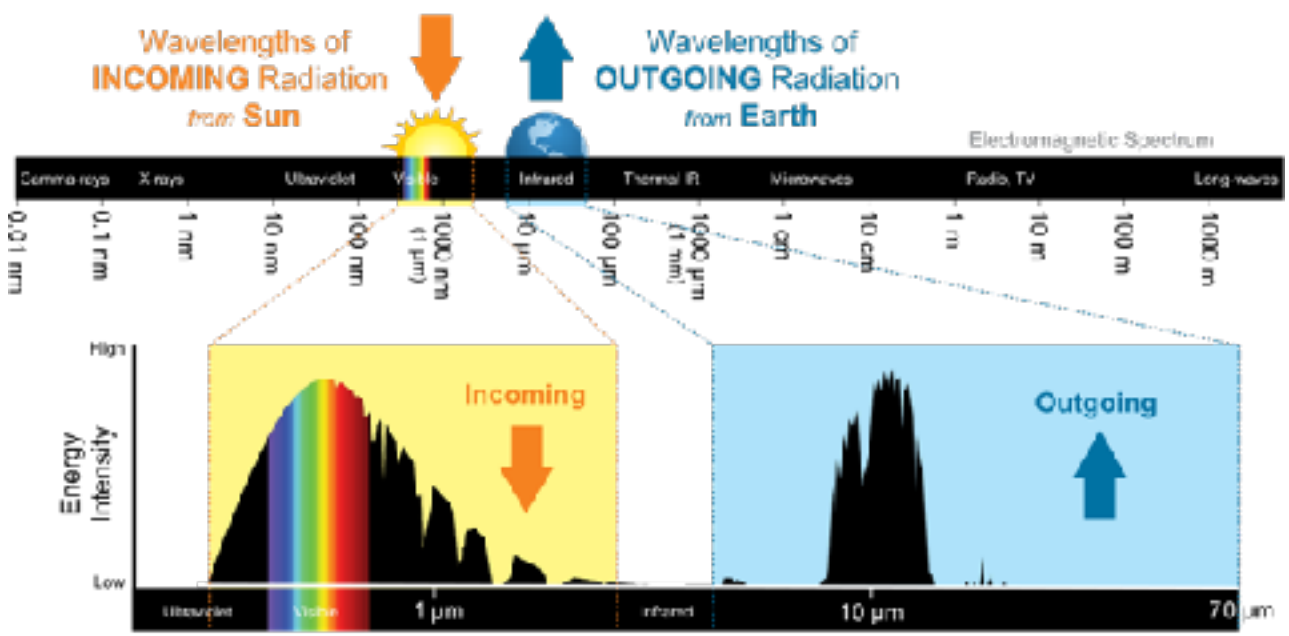
The amount Hydrocarbons employed to power our civilization is so large that it is significantly increasing the global concentration of CO_2 in the atmosphere.

The Temperature of Earth



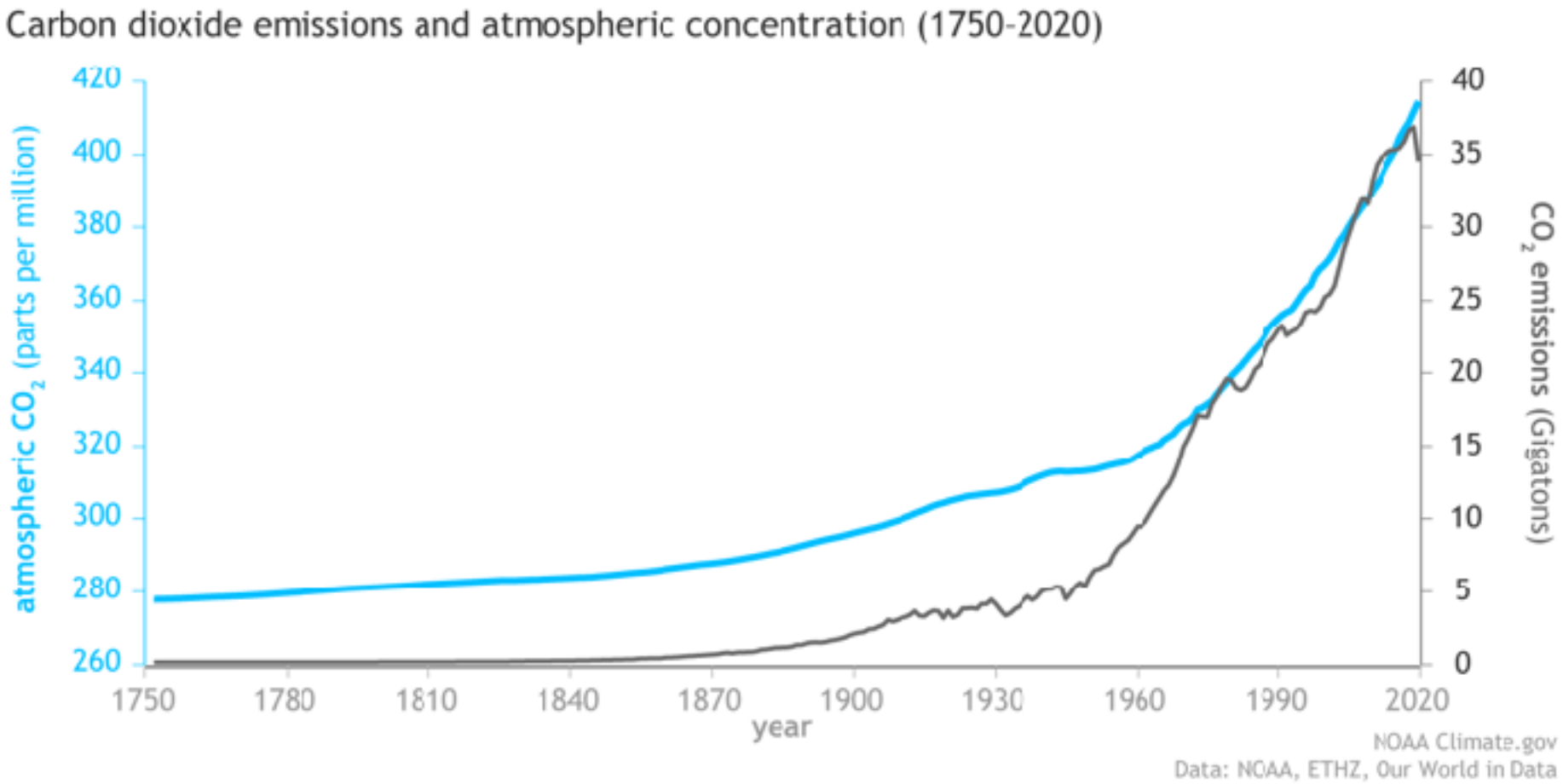
In absence of an atmosphere the average T of Earth would be -34ºC

The atmosphere is almost transparent to incident radiation but rather opaque to outgoing mainly due to H_2O and CO_2. This is the "Greenhouse Effect".



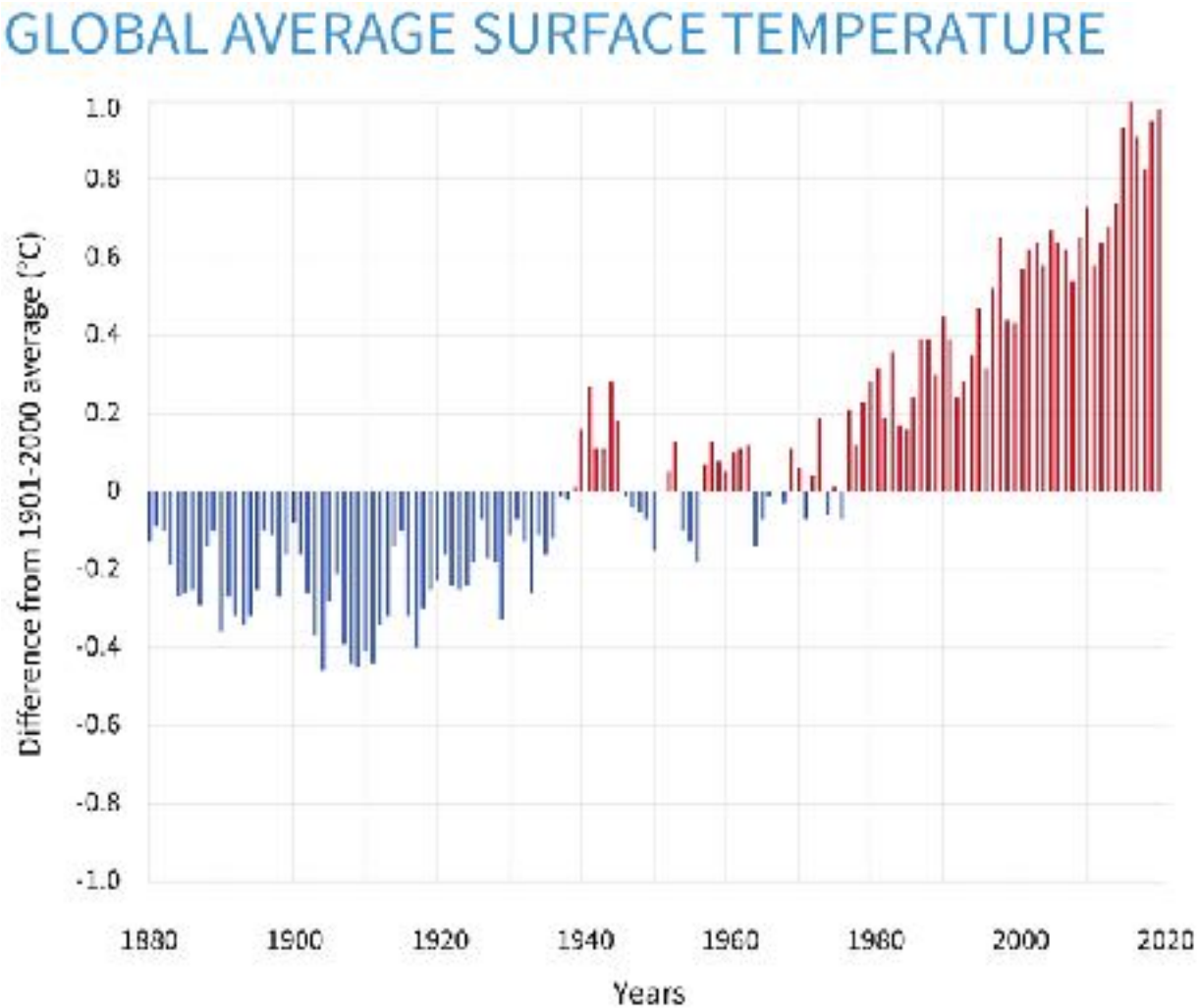
Temperature of the Earth

The atmospheric CO_2 concentration is rising due to global consumption of “fossil fuels”: Coal, Oil and Natural Gas.



Temperature of the Earth

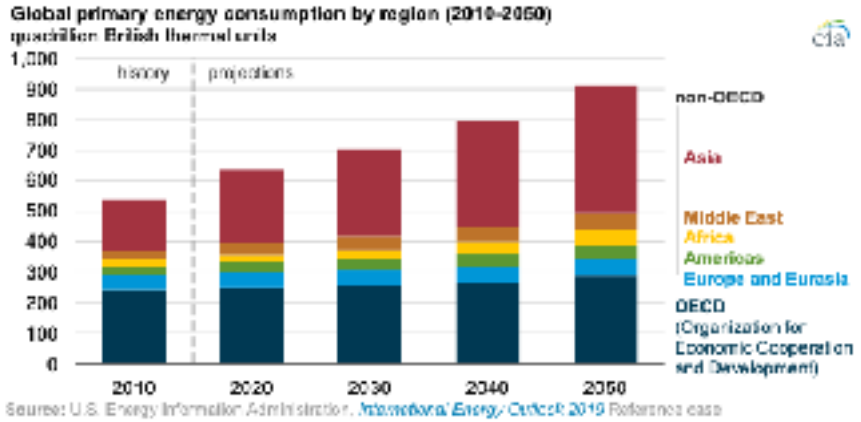
This more effective greenhouse effect causes the global temperature to increase.

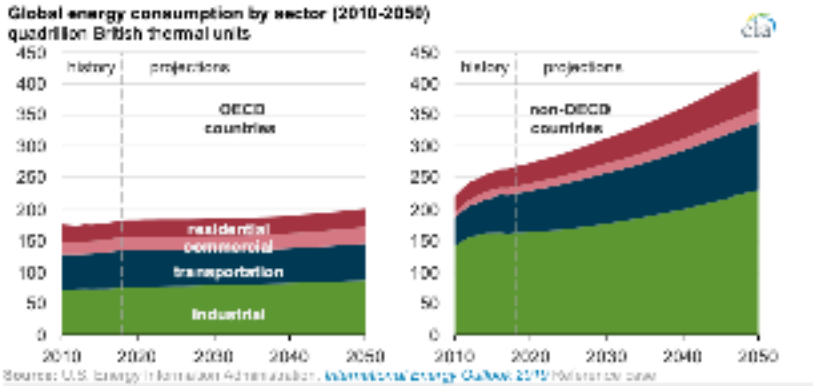


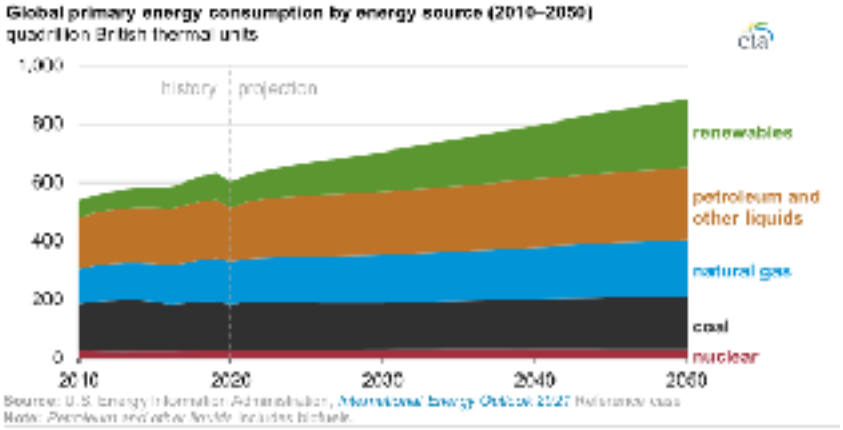
(source https://data.giss.nasa.gov/gistemp/)

Global climate change

- Large scale model shows that increasing the global temperature by more than 1.5 degrees will significantly change the Earth's climate.
- A 6 degree increase at the end of the last Ice Age melted the kilometre thick glaciers in the Northern Hemisphere.
- This requires stopping the use of fossil fuels for Power Production.

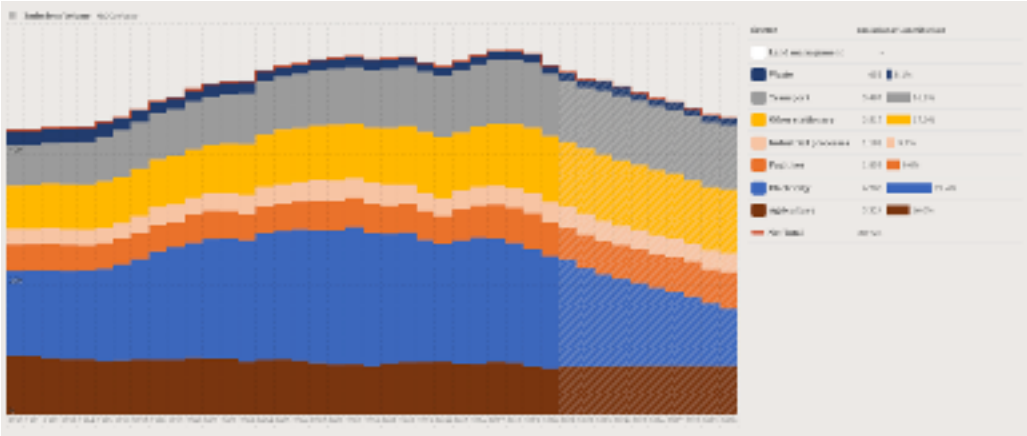


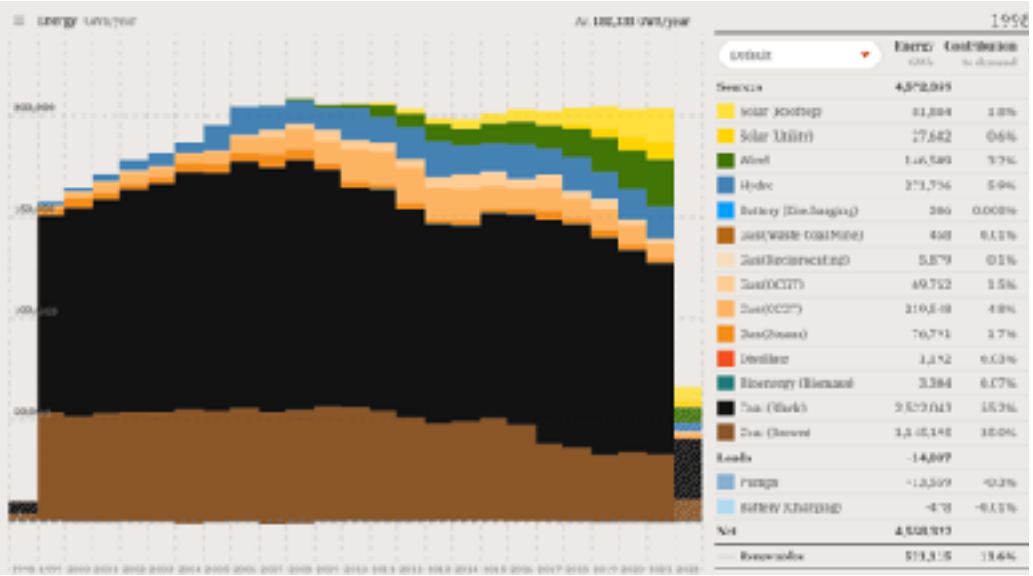




Emissions projections and electricity sources in Australia

Limiting the effects of climate change requires significantly reducing CO_2 emissions





Sources: OpenNEM, AEMO, APVI, BoM

The combination of electrification of the global economy and the use of non-CO_2 power sources shows the most promise

Summary

Nuclear Fusion:

$$ {}^{2}_{1}H + {}^{3}_{1}H \rightarrow {}^{4}_{2}He + n \quad (17.59\ MeV) $$

- Our civilisation delivers a historically unprecedented level of comfort
- Ultimately this is a large local reduction of entropy for humans
- Requires very large-scale use of cheap power
- This is delivered via heat-engines and electricity
- Current source of energy, fossil fuels drives Global Climate Change

- We'll continue this topic in the final lecture